

**Online Learning Management System with Performance Analytics for**  
**Senior High School Students**

A Capstone Project presented to the  
College of Computer Studies of  
Pamantasan ng Lungsod ng Pasig

In partial fulfillment of the requirements for the degree of  
Bachelor of Science in Information Technology

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## **ABSTRACT**

**Title** : Online Learning Management System with Performance Analytics for Senior High School Students

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**Degree** : Bachelor of Science in Information Technology

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**Project Description** : This study presents the development of a web-based Online Learning Management System (LMS) integrated with performance analytics to enhance the teaching and learning experience for senior high school students. Traditional classroom methods and basic online platforms often lack centralized management, real-time performance tracking, and detailed insights into student progress. The proposed system allows teachers to upload learning materials, create assessments, and monitor student performance, while students can access lessons, submit



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assignments, and track their academic progress. The system includes analytics features that evaluate student performance trends, identify learning gaps, and generate reports to support academic decision-making.

The system was developed using Agile methodology and evaluated based on ISO 25010 software quality standards, including functionality, usability, reliability, performance efficiency, security, maintainability, and portability. Results indicate that the system improves student engagement, supports data-driven instruction, and enhances overall learning outcomes.



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## **CHAPTER 1: INTRODUCTION**

### **I. Introduction**

Education has evolved significantly with the integration of digital technologies, especially in response to the growing demand for flexible and accessible learning environments. However, many schools still face challenges in managing online classes, tracking student performance, and maintaining organized learning resources.

Learning Management Systems (LMS) provide a centralized platform for delivering educational content, managing assessments, and monitoring student progress. When combined with performance analytics, these systems can offer valuable insights into student learning behavior and academic performance.

This study proposes an Online Learning Management System with performance analytics to improve teaching efficiency, enhance student engagement, and support data-driven educational practices.

### **II. Significance of the Study**

This system will benefit the following:

- **Students** – Easy access to learning materials and performance tracking
- **Teachers** – Efficient class management and assessment tools
- **School Administrators** – Centralized monitoring of academic performance
- **Educational Institutions** – Improved quality of education through data insights



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### **III. Statement of the Problem**

#### **General Problem**

How can an LMS with performance analytics improve teaching and learning outcomes for senior high school students?

#### **Specific Problems**

1. How to provide a centralized platform for learning materials and assessments?
2. How to monitor and evaluate student performance effectively?
3. How to identify learning gaps using performance data?
4. How to generate reports for academic monitoring and decision-making?

### **IV. Scope and Limitations**

#### **Scope**

- Online course and content management
- Assignment submission and assessment
- Performance analytics and reporting
- User roles: Admin, Teacher, Student

#### **Limitations**

- Requires stable internet access
- Limited to academic use within the institution



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- Does not include advanced AI-based tutoring features

## **OPERATIONAL DEFINITION OF TERMS**

**Learning Management System (LMS):** A web-based platform used to deliver, manage, and track educational content, activities, and student progress.

**Online Learning:** A mode of education where teaching and learning occur through the internet using digital platforms and tools.

**Performance Analytics:** The process of collecting, analyzing, and interpreting student data (e.g., grades, activity logs) to evaluate academic performance.

**Senior High School Students:** Learners enrolled in Grades 11 and 12 under the K-12 education system.

**Dashboard:** A visual interface within the system that displays key data such as student performance, progress, and statistics.

**User Interface (UI):** The visual components of the system that allow users to interact with the platform.

**User Experience (UX):** The overall experience of users when interacting with the system, including ease of use and satisfaction.

**Assessment:** Activities such as quizzes, exams, or assignments used to evaluate student learning.



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**Progress Tracking:** The monitoring of a student's completion of lessons, activities, and performance over time.

**Data Visualization:** The graphical representation of data (e.g., charts, graphs) to make information easier to understand.

**Student Performance:** The measurable academic achievement of students based on grades, scores, and participation.

**Instructor:** A teacher or facilitator who manages course content, assessments, and student progress in the system.

**Administrator:** A user responsible for managing system settings, users, and overall platform operation.

**Course Module:** A structured unit of learning content within a subject, including lessons, activities, and assessments.

**Real-Time Feedback:** Immediate responses provided by the system or instructor based on student performance.

**Database:** A structured collection of data used to store user information, course content, and performance records.

**Analytics Report:** A generated summary of student data used to evaluate trends and performance outcomes.